



		600	600	
		16	16	
		4	4	

Passed all tests! ✓

Question 3

Incorrect

Marked out of 7.00

Flag question

A bakery sells loaves of bread for \$3.49 each. Day old bread is discounted by 60 percent. Write a program that begins by reading the number of loaves of day old bread being purchased from the user. Then your program should display the regular price for the bread, the discount because it is a day old, and the total price. Each of these amounts should be displayed on its own line with an appropriate label. All of the values should be displayed using two decimal places.

Input Format

Read the number of day old loaves.

Output Format

First line, print Regular price: price

Second line, print Discount: discount

Third line, print Total: total

Note: All of the values should be displayed using two decimal places.

Sample Input 1

10

Sample Output 1

Regular price: 34.90

Discount: 20.94

Total: 13.96

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int num1;
5     scanf("%d",&num1);
6     float Regularprice= num1 *3.49;
7     float Discount= 0.6 * regularprice;
8     float Total= regularprice-discount;
9     printf("Regularprice :%.2f\n",regular
10    printf("Discount :%.2f\n",discount);
11    printf("Total :%.2f\n",total);
12    return 0;
13 }
```

Syntax Error(s)

```
__tester__.c: In function 'main':
__tester__.c:7:27: error: 'regularprice' undeclared (first
 7 |     float Discount= 0.6 * regularprice;
  |                               ^~~~~~
  |                               Regularprice
__tester__.c:7:27: note: each undeclared identifier is
__tester__.c:8:31: error: 'discount' undeclared (first
 8 |     float Total= regularprice-discount;
  |                               ^~~~~~
  |                               Discount
__tester__.c:11:28: error: 'total' undeclared (first
11 |     printf("Total :%.2f\n",total);
  |                               ^~~~~
  |                               Total
__tester__.c:8:11: error: unused variable 'Total'
 8 |     float Total= regularprice-discount;
  |     ^~~~~
__tester__.c:7:11: error: unused variable 'Discount'
 7 |     float Discount= 0.6 * regularprice;
  |     ^~~~~~
__tester__.c:6:11: error: unused variable 'Regularprice'
 6 |     float Regularprice= num1 *3.49;
  |     ^~~~~~
cc1: all warnings being treated as errors
```

Finish review



GE23131-Programming Using C-2024

Quiz navigation



Show one page at a time

Finish review

Question 1

Correct

Marked out of 3.00

Flag question

Write a program to read two integer values and print true if both the numbers end with the same digit, otherwise print false. Example: If 698 and 768 are given, program should print true as they both end with 8. Sample Input 1 25 53 Sample Output 1 false Sample Input 2 27 77 Sample Output 2 true

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int x,y;
5     scanf("%d%d",&x,&y);
6     if(x%10==y%10)
7     {
8         printf("true");
9     }
10    else
11    {
12        printf("false");
13    }
14    return 0;
15 }
```

	Input	Expected	Got	
✓	25 53	false	false	✓
✓	27 77	true	true	✓

Passed all tests! ✓

Question 2

Correct

Marked out of 5.00

Flag question

Objective

In this challenge, we're getting started with conditional statements.

Task

Given an integer, *n*, perform the following conditional actions:

- If *n* is odd, print *Weird*
- If *n* is even and in the inclusive range of 2 to 5, print *Not Weird*
- If *n* is even and in the inclusive range of 6 to 20, print *Weird*
- If *n* is even and greater than 20, print *Not Weird*

Complete the stub code provided in your editor to print whether or not *n* is weird.

Input Format

A single line containing a positive integer, *n*.

 Snapchat • Madhu • now 



sent you a Snap



View

Given an integer, *n*, perform the following conditional actions:

- If *n* is odd, print **Weird**
- If *n* is even and in the inclusive range of **2** to **5**, print **Not Weird**
- If *n* is even and in the inclusive range of **6** to **20**, print **Weird**
- If *n* is even and greater than **20**, print **Not Weird**

Complete the stub code provided in your editor to print whether or not *n* is weird.

Input Format

A single line containing a positive integer, *n*.

Constraints

- $1 \leq n \leq 100$

Output Format

Print **Weird** if the number is weird; otherwise, print **Not Weird**.

Sample Input 0

3

Sample Output 0

Weird

Sample Input 1

24

Sample Output 1

Not Weird

Explanation

Sample Case 0: n = 3
n is odd and odd numbers are weird, so we print **Weird**.

Sample Case 1: n = 24
n > 20 and *n* is even, so it isn't weird. Thus, we print **Not Weird**.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n;
```




Question **2**

Correct

Marked out of 5.00

Flag question

Objective

In this challenge, we're getting started with conditional statements.

Task

Given an integer, *n*, perform the following conditional actions:

- If *n* is odd, print Weird
- If *n* is even and in the inclusive range of 2 to 5, print **Not Weird**
- If *n* is even and in the inclusive range of 6 to 20, print **Weird**
- If *n* is even and greater than 20, print **Not Weird**

Complete the stub code provided in your editor to print whether or not *n* is weird.

Input Format

A single line containing a positive integer, *n*.

Constraints

- $1 \leq n \leq 100$

Output Format

Print Weird if the number is weird; otherwise, print Not Weird.

Sample Input 0

3

Sample Output 0

Weird

Sample Input 1

24

Sample Output 1

Not Weird

Explanation

Sample Case 0: *n* = 3
n is odd and odd numbers are weird, so we print **Weird**.

Sample Case 1: *n* = 24
n > 20 and *n* is even, so it isn't weird. Thus, we print **Not Weird**.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n;
5     scanf("%d",&n);
6     if(n%2!=0)
7     {
8         printf("Weird");
9     }
10    else if(n%2==0&&n>=2&&n<=5)
11    {
```


Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n;
5     scanf("%d",&n);
6     if(n%2!=0)
7     {
8         printf("Weird");
9     }
10    else if(n%2==0&&n>=2&&n<=5)
11    {
12        printf("Not Weird");
13    }
14    else if(n%2==0&&n>=6&&n<=20)
15    {
16        printf("Weird");
17    }
18    else if(n%2==0&&n>20)
19    {
20        printf("Not Weird");
21    }
22    else{
23        printf("Weird");
24    }
25 }
```

	Input	Expected	Got	
✓	3	Weird	Weird	✓
✓	24	Not Weird	Not Weird	✓

Passed all tests! ✓

Question 3

Correct

Marked out of 7.00

Flag question

Three numbers form a Pythagorean triple if the sum of squares of two numbers is equal to the square of the third. For example, 3, 5 and 4 form a Pythagorean triple, since $3^2 + 4^2 = 25 = 5^2$. You are given three integers, a, b, and c. They need not be given in increasing order. If they form a Pythagorean triple, then print "yes", otherwise, print "no". Please note that the output message is in small letters. Sample Input 1 3 5 4 Sample Output 1 yes Sample Input 2 5 8 2 Sample Output 2 no

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int a,b,c;
5     scanf("%d%d%d",&a,&b,&c);
6     if(a*a+b*b==c*c)
7     {
8         printf("yes");
9     }
10    else if(a*a+c*c==b*b)
11    {
12        printf("yes");
13    }
14    else if(b*b+c*c==a*a)
15    {
16        printf("yes");
17    }
18    else
19    {
20        printf("no");
21    }
22    return 0;
23 }
24
```

	Input	Expected	Got	
✓	3 5 4	yes	yes	✓
✓	5 8 2	no	no	✓

Passed all tests! ✓

Finish review



GE23131-Programming Using C-2024

Quiz navigation



Show one page at a time

Finish review

Question **1**

Correct

Marked out of 3.00

Flag question

Status	Finished
Started	Monday, 23 December 2024, 5:33 PM
Completed	Wednesday, 4 December 2024, 11:49 AM
Duration	19 days 5 hours

Write a program that determines the name of a shape from its number of sides. Read the number of sides from the user and then report the appropriate name as part of a meaningful message. Your program should support shapes with anywhere from 3 up to (and including) 10 sides. If a number of sides outside of this range is entered then your program should display an appropriate error message.

Sample Input 1

3

Sample Output 1

Triangle

Sample Input 2

7

Sample Output 2

Heptagon

Sample Input 3

11

Sample Output 3

The number of sides is not supported.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n;
5     scanf("%d",&n);
6     if(n==3)
7     {
8         printf("Triangle");
9     }
10    else if(n==4)
11    {
12        printf("Square");
13    }
14    else if(n==5)
15    {
16        printf("Pentagon");
17    }
18    else if(n==6)
19    {
20        printf("Hexagon");
21    }
22    else if(n==7)
23    {
24        printf("Heptagon");
25    }
26    else if(n==8)
27    {
28        printf("Octagon");
29    }
```




REC-CIS

Answer: (penalty regime: 0 %)

```
3 {
4     int n;
5     scanf("%d",&n);
6     if(n==3)
7     {
8         printf("Triangle");
9     }
10    else if(n==4)
11    {
12        printf("Square");
13    }
14    else if(n==5)
15    {
16        printf("Pentagon");
17    }
18    else if(n==6)
19    {
20        printf("Hexagon");
21    }
22    else if(n==7)
23    {
24        printf("Heptagon");
25    }
26    else if(n==8)
27    {
28        printf("Octagon");
29    }
30    else if(n==9)
31    {
32        printf("Nonagon");
33    }
34    else if(n==10)
35    {
36        printf("Decagon");
37    }
38    else
39    {
40        printf("The number of sides is not supported.");
41    }
42 }
43
44
45
46
47
48
49
50
51
52
53
54
```

	Input	Expected
✓	3	Triangle
✓	7	Heptagon
✓	11	The number of sides is not supported.

Passed all tests! ✓

Question **2**

Correct

Marked out of 5.00

Flag question

The Chinese zodiac assigns animals to years in a 12-year cycle. One 12-year cycle is shown in the table below. The pattern repeats from there, with 2012 being another year of the Dragon, and 1999 being another year of the Hare.

Year	Animal
2000	Dragon
2001	Snake
2002	Horse
2003	Sheep
2004	Monkey
2005	Rooster
2006	Dog
2007	Pig

Question **2**

Correct

Marked out of 5.00

Flag question

The Chinese zodiac assigns animals to years in a 12-year cycle. One 12-year cycle is shown in the table below. The pattern repeats from there, with 2012 being another year of the Dragon, and 1999 being another year of the Hare.

Year	Animal
2000	Dragon
2001	Snake
2002	Horse
2003	Sheep
2004	Monkey
2005	Rooster
2006	Dog
2007	Pig
2008	Rat
2009	Ox
2010	Tiger
2011	Hare

Write a program that reads a year from the user and displays the animal associated with that year. Your program should work correctly for any year greater than or equal to zero, not just the ones listed in the table.

Sample Input 1

2004

Sample Output 1

Monkey

Sample Input 2

2010

Sample Output 2

Tiger

Answer: (penalty regime: 0 %)

```
4 int year;
5 scanf("%d",&year);
6 if(year%12==8)
7 {
8     printf("Dragon");
9 }
10 else if(year%12==9)
11 {
12     printf("Snake");
13 }
14 else if(year%12==10)
15 {
16     printf("Horse");
17 }
18 else if(year%12==11)
19 {
20     printf("Sheep");
21 }
22 else if(year%12==0)
23 {
24     printf("Monkey");
25 }
26 else if(year%12==1)
27 {
28     printf("Rooster");
29 }
30 else if(year%12==2)
31 {
32     printf("Dog");
33 }
34 else if(year%12==3)
35 {
36     printf("Pig");
37 }
38 else if(year%12==4)
39 {
40     printf("Rat");
```


Answer. (penalty regime: 0 %)

```
4 int year;
5 scanf("%d",&year);
6 if(year%12==8)
7 {
8     printf("Dragon");
9 }
10 else if(year%12==9)
11 {
12     printf("Snake");
13 }
14 else if(year%12==10)
15 {
16     printf("Horse");
17 }
18 else if(year%12==11)
19 {
20     printf("Sheep");
21 }
22 else if(year%12==0)
23 {
24     printf("Monkey");
25 }
26 else if(year%12==1)
27 {
28     printf("Rooster");
29 }
30 else if(year%12==2)
31 {
32     printf("Dog");
33 }
34 else if(year%12==3)
35 {
36     printf("Pig");
37 }
38 else if(year%12==4)
39 {
40     printf("Rat");
41 }
42 else if(year%12==5)
43 {
44     printf("Ox");
45 }
46 else if(year%12==6)
47 {
48     printf("Tiger");
49 }
50 else
51 {
52     printf("Hare");
53 }
54 return 0;
55 }
```

	Input	Expected	Got	
✓	2004	Monkey	Monkey	✓
✓	2010	Tiger	Tiger	✓

Passed all tests! ✓

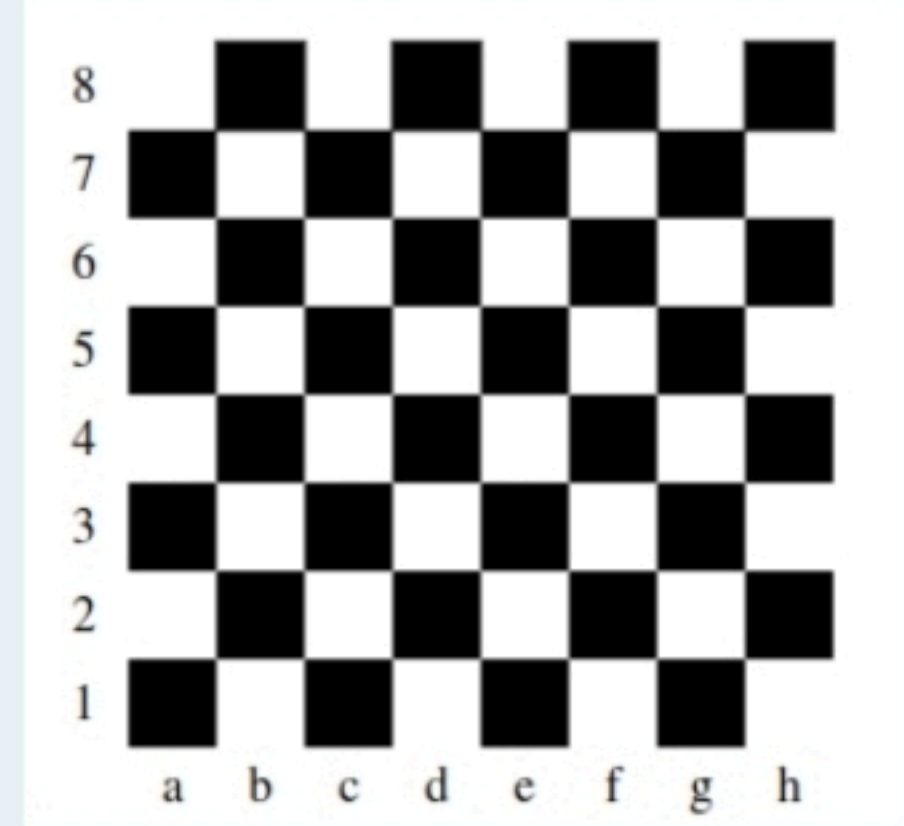
Question **3**

Correct

Marked out of 7.00

 Flag question

Positions on a chess board are identified by a letter and a number. The letter identifies the column, while the number identifies the row, as shown below:



Write a program that reads a position from the user. Use an if statement to determine if the column begins with a black square or a white square. Then use modular arithmetic to report the color of the square in that row. For example, if the user enters a1 then your program should report that the square is black. If the user enters d5 then your program should report that the square is white. Your program may assume that a valid position will always be entered. It does not need to perform any error checking.

Sample Input 1



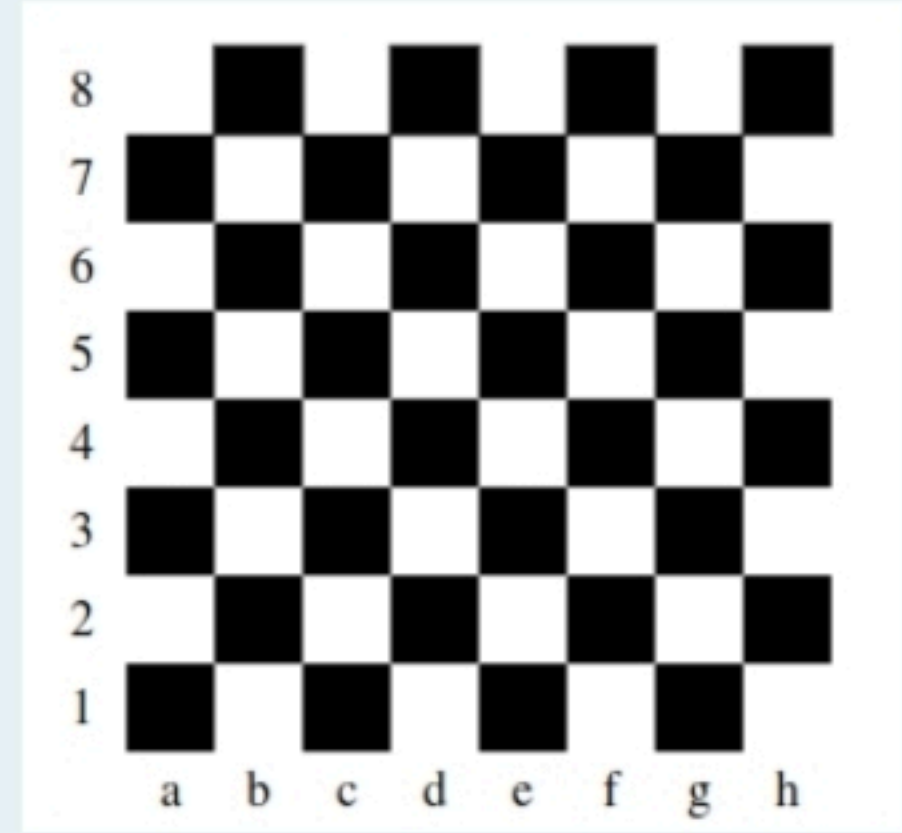
Question 3

Correct

Marked out of 7.00

Flag question

Positions on a chess board are identified by a letter and a number. The letter identifies the column, while the number identifies the row, as shown below:



Write a program that reads a position from the user. Use an if statement to determine if the column begins with a black square or a white square. Then use modular arithmetic to report the color of the square in that row. For example, if the user enters a1 then your program should report that the square is black. If the user enters d5 then your program should report that the square is white. Your program may assume that a valid position will always be entered. It does not need to perform any error checking.

Sample Input 1

a 1

Sample Output 1

The square is black.

Sample Input 2

d 5

Sample Output 2

The square is white.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int num,sum;
5     char alpha;
6     scanf("%c%d",&alpha,&num);
7     sum=alpha+num;
8     if(sum%2==0)
9     {
10         printf("The square is black.");
11     }
12     else
13     {
14         printf("The square is white.");
15     }
16 }
```

	Input	Expected	Got
✓	a 1	The square is black.	The square is bl
✓	d 5	The square is white.	The square is wh

Passed all tests! ✓

Finish review



GE23131-Programming Using C-2024

Quiz navigation

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- 3

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Finish review

Question 1

Correct

Marked out of 3.00

Flag question

Status	Finished
Started	Monday, 23 December 2024, 5:33 PM
Completed	Monday, 9 December 2024, 11:23 AM
Duration	14 days 6 hours

Some data sets specify dates using the year and day of year rather than the year, month, and day of month. The day of year (DOY) is the sequential day number starting with day 1 on January 1st.

There are two calendars - one for normal years with 365 days, and one for leap years with 366 days. Leap years are divisible by 4. Centuries, like 1900, are not leap years unless they are divisible by 400. So, 2000 was a leap year.

To find the day of year number for a standard date, scan down the Jan column to find the day of month, then scan across to the appropriate month column and read the day of year number. Reverse the process to find the standard date for a given day of year.

Write a program to print the Day of Year of a given date, month and year.

Sample Input 1

18
6
2020

Sample Output 1

170

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int i,mon,yr,inday;
5     scanf("%d%d%d",&inday,&mon,&yr);
6     char daypm[]={31,28,31,30,31,30,31,30,31,31,30,31};
7     int day=inday;
8     if((yr%4==0&&yr%100!=0)|| (yr%400==0))
9     {
10         daypm[1]=29;
11     }
12     for(i=0;i<mon-1;i++)
13     {
14         day=day+daypm[i];
15     }
16     printf("%d",day);
17     return 0;
18 }
19 }
```

	Input	Expected	Got	
✓	18 6 2020	170	170	✓

Passed all tests! ✓



Question **2**

Correct

Marked out of 5.00

Flag question

Suppandi is trying to take part in the local village math quiz. In the first round, he is asked about shapes and areas. Suppandi, is confused, he was never any good at math. And also, he is bad at remembering the names of shapes. Instead, you will be helping him **calculate the area** of shapes.

- When he says rectangle he is actually referring to a square.
- When he says square, he is actually referring to a triangle.
- When he says triangle he is referring to a rectangle
- And when he is confused, he just says something random. At this point, all you can do is say 0.

Help Suppandi by printing the correct answer in an integer.

Input Format

- Name of shape (always in upper case R à Rectangle, S à Square, T à Triangle)
- Length of 1 side
- Length of other side

Note: In case of triangle, you can consider the sides as height and length of base

Output Format

- Print the area of the shape.

Sample Input 1

T
10
20

Sample Output 1

200

Sample Input 2

S
30
40

Sample Output 2

600

Sample Input 3

R
10
10

Sample Output 3

100

Sample Input 4

C

Sample Input

C
9
10

Sample Output 4

0

Explanation:

- First is output of area of rectangle
- Then, output of area of triangle
- Then output of area square
- Finally, something random, so we print 0

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int a,b;
5     char c;
6     scanf("%c%d%d",&c,&a,&b);
7     switch(c)
8     {
9         case'R':
10            printf("%d",a*b);
11            break;
12         case'S':
13            printf("%.0f",a*b*(0.5));
14            break;
15         case'T':
16            printf("%d",a*b);
17            break;
18         default:
19            printf("0");
20     }
21     return 0;
22 }
```

	Input	Expected	Got	
✓	T 10 20	200	200	✓
✓	S 30 40	600	600	✓
✓	B 2 11	0	0	✓
✓	R 10 30	300	300	✓
✓	S 40 50	1000	1000	✓

Passed all tests! ✓

Question 3

Correct

Marked out of 7.00

 Flag question

Superman is planning a journey to his home planet. It is very important for him to know which day he arrives there. They don't follow the 7-day week like us. Instead, they follow a 10-day week with the following days: Day Number Name of Day
1 Sunday 2 Monday 3 Tuesday 4 Wednesday 5 Thursday 6 Friday 7 Saturday 8 Kryptonday 9 Coluday 10 Daxamday
Here are the rules of the calendar: • The calendar starts with Sunday always. • It has only 296 days. After the 296th day, it goes back to Sunday. You begin your journey on a Sunday and will reach after n. You have to tell on which day you will arrive when you reach there.

Input format: •

Contain a number n (0 < n)

Output format: Print the name of the day you are arriving on

Example Input

Passed all tests! ✓

Question 3

Correct

Marked out of 7.00

 Flag question

Superman is planning a journey to his home planet. It is very important for him to know which day he arrives there. They don't follow the 7-day week like us. Instead, they follow a 10-day week with the following days: Day Number Name of Day
1 Sunday 2 Monday 3 Tuesday 4 Wednesday 5 Thursday 6 Friday 7 Saturday 8 Kryptonday 9 Coluday 10 Daxamday
Here are the rules of the calendar: • The calendar starts with Sunday always. • It has only 296 days. After the 296th day, it goes back to Sunday. You begin your journey on a Sunday and will reach after n. You have to tell on which day you will arrive when you reach there.

Input format: •

Contain a number n ($0 < n$)

Output format: Print the name of the day you are arriving on

Example Input

7

Example Output

Kryptonday

Example Input

1

Example Output Monday

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,day;
5     scanf("%d",&n);
6     if(n<296)
7         day=n;
8     else
9         day=n-296;
10    day%=10;
11    day=day+1;
12    day%=10;
13    switch(day)
14    {
15        case 1:
16            printf("Sunday");
17            break;
18        case 2:
19            printf("Monday");
20            break;
21        case 3:
22            printf("Tuesday");
23            break;
24        case 4:
25            printf("Wednesday");
26            break;
27        case 5:
28            printf("Thursday");
29            break;
30        case 6:
31            printf("Friday");
32            break;
33        case 7:
34            printf("Saturday");
35            break;
36        case 8:
37            printf("Kryptonday");
38            break;
39        case 9:
40            printf("Coluday");
41            break;
42        case 10:
43            printf("Daxamday");
44            break;
45    }
46 }
```

	Input	Expected	Got	
✓	7	Kryptonday	Kryptonday	✓
✓	1	Monday	Monday	✓

Passed all tests! ✓

Finish review



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Quiz navigation



Show one page at a time

Finish review

Status	Finished
Started	Monday, 23 December 2024, 5:33 PM
Completed	Wednesday, 11 December 2024, 11:37 AM
Duration	12 days 5 hours

Question 1

Correct

Marked out of 3.00

Flag question

Alice and Bob are playing a game called "Stone Game". Stone game is a two-player game. Let N be the total number of stones. In each turn, a player can remove either one stone or four stones. The player who picks the last stone, wins. They follow the "Ladies First" norm. Hence Alice is always the one to make the first move. Your task is to find out whether Alice can win, if both play the game optimally.

Input Format

First line starts with T, which is the number of test cases. Each test case will contain N number of stones.

Output Format

Print "Yes" in the case Alice wins, else print "No".

Constraints

1<=T<=1000

1<=N<=10000

Sample Input and Output

Input

3
1
6
7

Output

Yes
Yes
No

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int T,i=0,n,t;
5     scanf("%d",&T);
6     while(i<T)
7     {
8         scanf("%d",&n);
9         t=n/4;
10        if(t%2==0 && n%2==0)
11        {
12            printf("No\n");
13        }
14        else if(t%2==1&&n%2==1)
15        {
16            printf("No\n");
17        }
18        else
```

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int T,i=0,n,t;
5     scanf("%d",&T);
6     while(i<T)
7     {
8         scanf("%d",&n);
9         t=n/4;
10        if(t%2==0 && n%2==0)
11        {
12            printf("No\n");
13        }
14        else if(t%2==1&&n%2==1)
15        {
16            printf("No\n");
17        }
18        else
19        {
20            printf("Yes\n");
21        }
22        i++;
23    }
24    return 0;
25 }
26
27
```

	Input	Expected	Got	
✓	3	Yes	Yes	✓
	1	Yes	Yes	
	6	No	No	
	7			

Passed all tests! ✓

Question **2**

Correct

Marked out of 5.00

Flag question

You are designing a poster which prints out numbers with a unique style applied to each of them. The styling is based on the number of closed paths or holes present in a given number.

The number of holes that each of the digits from 0 to 9 have are equal to the number of closed paths in the digit. Their values are:

- 1, 2, 3, 5, and 7 = 0 holes.
- 0, 4, 6, and 9 = 1 hole.
- 8 = 2 holes.

Given a number, you must determine the sum of the number of holes for all of its digits. For example, the number 819 has 3 holes.

Complete the program, it must must return an integer denoting the total number of holes in num.

Constraints

$1 \leq \text{num} \leq 109$

Input Format For Custom Testing

There is one line of text containing a single integer num, the value to process.

Sample Input

630

Sample Output

2

Explanation



REC-CIS

Add the holes count for each digit, 1, 2, 8, 8. Return 0 + 0 + 2 + 2 = 4.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int a,b,n=0;
5     scanf("%d",&a);
6     while(a>0)
7     {
8         b=a%10;
9         if(b==0 || b==6 || b==9 || b==4)
10        {
11            n=n+1;
12        }
13        else if(b==8)
14        {
15            n=n+2;
16        }
17        a=a/10;
18    }
19    printf("%d",n);
20    return 0;
21 }
22
23 }
```

	Input	Expected	Got	
✓	630	2	2	✓
✓	1288	4	4	✓

Passed all tests! ✓

Question 3

Correct

Marked out of 7.00

Flag question

The problem solvers have found a new Island for coding and named it as Philaland. These smart people were given a task to make a purchase of items at the Island easier by distributing various coins with different values. Manish has come up with a solution that if we make coins category starting from \$1 till the maximum price of the item present on Island, then we can purchase any item easily. He added the following example to prove his point.

Let's suppose the maximum price of an item is 5\$ then we can make coins of { \$1, \$2, \$3, \$4, \$5 } to purchase any item ranging from \$1 till \$5.

Now Manisha, being a keen observer suggested that we could actually minimize the number of coins required and gave following distribution { \$1, \$2, \$3 }. According to him any item can be purchased one time ranging from \$1 to \$5. Everyone was impressed with both of them. Your task is to help Manisha come up with a minimum number of denominations for any arbitrary max price in Philaland.

Input Format

Contains an integer N denoting the maximum price of the item present on Philaland.

Output Format

Print a single line denoting the minimum number of denominations of coins required.

Constraints

1<=T<=100

10

Sample Output 1:

4

Sample Input 2:

5

Sample Output 2:

3

Explanation:

For test case 1, N=10.

According to Manish {\$1, \$2, \$3,... \$10} must be distributed.

But as per Manisha only {\$1, \$2, \$3, \$4} coins are enough to purchase any item ranging from \$1 to \$10. Hence minimum is 4. Likewise denominations could also be {\$1, \$2, \$3, \$5}. Hence answer is still 4.

For test case 2, N=5.

According to Manish {\$1, \$2, \$3, \$4, \$5} must be distributed.

But as per Manisha only {\$1, \$2, \$3} coins are enough to purchase any item ranging from \$1 to \$5. Hence minimum is 3. Likewise, denominations could also be {\$1, \$2, \$4}. Hence answer is still 3.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,r=0;
5     scanf("%d",&n);
6     while(n!=0)
7     {
8         n=n/2;
9         r=r+1;
10    }
11    printf("%d",r);
12 }
```

	Input	Expected	Got	
✓	10	4	4	✓
✓	5	3	3	✓
✓	20	5	5	✓
✓	500	9	9	✓
✓	1000	10	10	✓

Passed all tests! ✓

Finish review



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Status	Finished
Started	Monday, 23 December 2024, 5:33 PM
Completed	Wednesday, 11 December 2024, 11:52 AM
Duration	12 days 5 hours

Question **1**

Correct

Marked out of 3.00

Flag question

A set of N numbers (separated by one space) is passed as input to the program. The program must identify the count of numbers where the number is odd number.

Input Format:

The first line will contain the N numbers separated by one space.

Boundary Conditions:

3 <= N <= 50

The value of the numbers can be from -99999999 to 99999999

Output Format:

The count of numbers where the numbers are odd numbers.

Example Input / Output 1:

Input:

5 10 15 20 25 30 35 40 45 50

Output:

5

Explanation:

The numbers meeting the criteria are 5, 15, 25, 35, 45.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,x=0;
5     while(scanf("%d",&n)==1)
6     {
7         if(n%2!=0)
8         {
9             x++;
10        }
11    }
12    printf("%d",x);
13    return 0;
14 }
```

The numbers meeting the criteria are 5, 15, 25, 35, 45.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,x=0;
5     while(scanf("%d",&n)==1)
6     {
7         if(n%2!=0)
8         {
9             x++;
10        }
11    }
12    printf("%d",x);
13    return 0;
14 }
```

	Input	Expected	Got
✓	5 10 15 20 25 30 35 40 45 50	5	5

Passed all tests! ✓

Question **2**

Correct

Marked out of 5.00

[Flag question](#)

Given a number N, return true if and only if it is a *confusing number*, which satisfies the following condition:

We can rotate digits by 180 degrees to form new digits. When 0, 1, 6, 8, 9 are rotated 180 degrees, they become 0, 1, 9, 8, 6 respectively. When 2, 3, 4, 5 and 7 are rotated 180 degrees, they become invalid. A *confusing number* is a number that when rotated 180 degrees becomes a **different** number with each digit valid.

Example 1:

6 -> 9

Input: 6

Output: true

Explanation:

We get 9 after rotating 6, 9 is a valid number and 9!=6.

Example 2:

89 -> 68

Input: 89

Output: true

Explanation:

We get 68 after rotating 89, 86 is a valid number and 86!=89.

Example 3:

11 -> 11

Input: 11

Output: false

Explanation:

We get 11 after rotating 11, 11 is a valid number but the value remains the same, thus 11 is not a confusing number.

Note:

- 0 <= N <= 10^9
- After the rotation we can ignore leading zeros, for example if after rotation we have 0008 then this number is considered as just 8.

Answer: (penalty regime: 0 %)

Example 2:

89 -> 68

Input: 89

Output: true

Explanation:

We get 68 after rotating 89, 86 is a valid number and 86!=89.

Example 3:

11 -> 11

Input: 11

Output: false

Explanation:

We get 11 after rotating 11, 11 is a valid number but the value remains the same, thus 11 is not a confusing number.

Note:

- 0 <= N <= 10^9
- After the rotation we can ignore leading zeros, for example if after rotation we have 0008 then this number is considered as just 8.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,x,y=1;
5     scanf("%d",&n);
6     while(n!=0 && y==1)
7     {
8         x=n%10;
9         n=n/10;
10        if((x==2) || (x==3) || (x==4) || (x==7)
11        {
12            y++;
13        }
14    }
15    if(y==1)
16    {
17        printf("true");
18    }
19    else
20    {
21        printf("false");
22    }
23    return 0;
24 }
```

	Input	Expected	Got	
✓	6	true	true	✓
✓	89	true	true	✓
✓	25	false	false	✓

Passed all tests! ✓

Question 3

Correct

Marked out of 7.00

Flag question

A nutritionist is labeling all the best power foods in the market. Every food item arranged in a single line, will have a value beginning from 1 and increasing by 1 for each, until all items have a value associated with them. An item's value is the same as the number of macronutrients it has. For example, food item with value 1 has 1 macronutrient, food item with value 2 has 2 macronutrients, and incrementing in this fashion.

The nutritionist has to recommend the best combination to patients, i.e. maximum total of macronutrients. However, the nutritionist must avoid prescribing a particular sum of macronutrients (an 'unhealthy' number), and this sum is known. The nutritionist chooses food items in the increasing order of their value. Compute the highest total of macronutrients that can be prescribed to a patient, without the sum matching the given 'unhealthy' number.

Here's an illustration:

2. $1 + 2 = 3$; observe that this is the max total, and having avoided having exactly $k = 2$ macronutrients.

Sample Input 1

2
1

Sample Output 1

2

Explanation 1

1. Cannot use item 1 because $k = 1$ and $sum \equiv k$ has to be avoided at any time.
2. Hence, max total is achieved by $sum = 0 + 2 = 2$.

Sample Case 2

Sample Input For Custom Testing

Sample Input 2

3
3

Sample Output 2

5

Explanation 2

$2 + 3 = 5$, is the best case for maximum nutrients.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     long long int n,t,i,nut=0;
5     scanf("%lld%lld",&n,&t);
6     for(i=1;i<=n;i++)
7     {
8         nut=nut+i;
9         if(nut==t)
10        {
11            nut=nut-1;
12        }
13    }
14    printf("%lld",nut%1000000007);
15 }
```

	Input	Expected	Got	
✓	2 2	3	3	✓
✓	2 1	2	2	✓
✓	3 3	5	5	✓

Passed all tests! ✓

Finish review



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Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int T,size;
5     scanf("%d",&T);
6     for(int t=0;t<T;t++)
7     {
8         scanf("%d",&size);
9         for(int i=0;i<size;i++)
10        {
11            for(int j=0;j<size;j++)
12            {
13                if((i+j)%2==0)
14                {
15                    printf("W");
16                }
17                else
18                {
19                    printf("B");
20                }
21            }
22            printf("\n");
23        }
24    }
25 }
26 }
```

	Input	Expected	Got	
✓	2	WBW	WBW	✓
	3	BWB	BWB	
	5	WBW	WBW	
		WBWBW	WBWBW	
		BWBWB	BWBWB	
		WBWBW	WBWBW	
		BWBWB	BWBWB	
		WBWBW	WBWBW	

Passed all tests! ✓

Question 2

Correct

Marked out of 5.00

Flag question

Let's print a chessboard!

Write a program that takes input:

The first line contains T, the number of test cases
Each test case contains an integer N and also the starting character of the chessboard

Output Format

Print the chessboard as per the given examples

Sample Input / Output

Input:

2
2 W
3 B

Output:

WB
BW
BWB
WBW
BWB

Answer: (penalty regime: 0 %)



REC-CIS

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Question 1

Correct

Marked out of 3.00

Flag question

Status	Finished
Started	Monday, 23 December 2024, 5:33 PM
Completed	Wednesday, 18 December 2024, 11:29 AM
Duration	5 days 6 hours

Write a program that prints a simple chessboard.

Input format:

The first line contains the number of inputs T.
The lines after that contain a different values for size of the chessboard

Output format:

Print a chessboard of dimensions size * size. Print a Print W for white spaces and B for black spaces.

Input:

2
3
5

Output:

WBW
BWB
WBW
WBWBW
BWBWB
WBWBW
BWBWB
WBWBW

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int T,size;
5     scanf("%d",&T);
6     for(int t=0;t<T;t++)
7     {
8         scanf("%d",&size);
9         for(int i=0;i<size;i++)
10        {
11            for(int j=0;j<size;j++)
12            {
13                if((i+j)%2==0)
14                {
15                    printf("W");
16                }
17                else
18                {
19                    printf("B");
20                }
21            }
22            printf("\n");
23        }
24    }
25 }
26 }
```


BWB
WBW
BWB

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int T,d,i,i1,i2,o,z;
5     char c,s;
6     scanf("%d",&T);
7     for(i=0;i<T;i++)
8     {
9         scanf("%d %c",&d,&s);
10        for(i1=0;i1<d;i1++)
11        {
12            z=(s=='W') ? 0:1;
13            o=(i1%2==z) ? 0:1;
14            for(i2=0;i2<d;i2++)
15            {
16                c=(i2%2==o) ? 'W':'B';
17                printf("%c",c);
18            }
19            printf("\n");
20        }
21    }
22    return 0;
23 }
```

	Input	Expected	Got	
✓	2	WB	WB	✓
	2 W	BW	BW	
	3 B	BWB	BWB	
		WBW	WBW	
		BWB	BWB	

Passed all tests! ✓

Question **3**
Correct
Marked out of 7.00
[Flag question](#)

Decode the logic and print the Pattern that corresponds to given input.

If N= 3

then pattern will be :

10203010011012
**4050809
****607

If N= 4, then pattern will be:

1020304017018019020
**50607014015016
****809012013
*****10011

Constraints

2 <= N <= 100

Input Format

First line contains T, the number of test cases
Each test case contains a single integer N

Output

First line print Case #i where i is the test case number
In the subsequent line, print the pattern

Test Case 1

REC-CIS

Case #1
10203010011012
**4050809
****607
Case #2
1020304017018019020
**50607014015016
****809012013
*****10011
Case #3
102030405026027028029030
**6070809022023024025
****10011012019020021
*****13014017018
*****15016

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,v,p3,c,in,i,i1,i2,t,ti;
5     scanf("%d",&t);
6     for(ti=0;ti<t;ti++)
7     {
8         v=0;
9         scanf("%d",&n);
10        printf("Case #%d\n",ti+1);
11        for(i=0;i<n;i++)
12        {
13            c=0;
14            if(i>0){
15                for(i1=0;i1<i;i1++)
16                    printf("**");
17            }
18            for(i1=i;i1<n;i1++)
19            {
20                if(i>0)
21                    c++;
22                printf("%d0",++v);
23            }
24            if(i==0){
25                p3=v+(v*(v-1))+1;
26                in=p3;
27            }
28            in=in-c;
29            p3=in;
30            for(i2=i;i2<n;i2++)
31            {
32                printf("%d",p3++);
33                if(i2!=n-1)
34                    printf("0");
35            }
36            printf("\n");
37        }
38    }
39
40 }
41 return 0;
42 }
```

	Input	Expected	Got
✓	3	Case #1	Case #1
	3	10203010011012	10203010011012
	4	**4050809	**4050809
	5	****607	****607
		Case #2	Case #2
		1020304017018019020	1020304017018019020
		**50607014015016	**50607014015016
		****809012013	****809012013
		*****10011	*****10011
		Case #3	Case #3
		102030405026027028029030	102030405026027028029030
		**6070809022023024025	**6070809022023024025
		****10011012019020021	****10011012019020021
		*****13014017018	*****13014017018
		*****15016	*****15016

Passed all tests! ✓

Finish review

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Finish review

Status	Finished
Started	Monday, 23 December 2024, 5:33 PM
Completed	Saturday, 21 December 2024, 10:46 AM
Duration	2 days 6 hours

Question **1**
Correct
Marked out of 3.00
 Flag question

The k-digit number N is an Armstrong number if and only if the k-th power of each digit sums to N.

Given a positive integer N, return true if and only if it is an Armstrong number.

Example 1:

Input:

153

Output:

true

Explanation:

153 is a 3-digit number, and $153 = 1^3 + 5^3 + 3^3$.

Example 2:

Input:

123

Output:

false

Explanation:

123 is a 3-digit number, and $123 \neq 1^3 + 2^3 + 3^3 = 36$.

Example 3:

Input:

1634

Output:

true

Note:

$1 \leq N \leq 10^8$

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 #include<math.h>
```



Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 #include<math.h>
3 int main()
4 {
5     int n;
6     scanf("%d",&n);
7     int x=0,n2=n;
8     while(n2!=0)
9     {
10         x++;
11         n2=n2/10;
12     }
13     int sum=0;
14     int n3=n,n4;
15     while(n3!=0)
16     {
17         n4=n3%10;
18         sum=sum+pow(n4,x);
19         n3=n3/10;
20     }
21     if(n==sum)
22     {
23         printf("true");
24     }
25     else
26     {
27         printf("false");
28     }
29     return 0;
30 }
```

	Input	Expected	Got	
✓	153	true	true	✓
✓	123	false	false	✓

Passed all tests! ✓

Question 2

Correct

Marked out of 5.00

Flag question

Take a number, reverse it and add it to the original number until the obtained number is a palindrome. Constraints 1<=num<=999999999 Sample Input 1 32 Sample Output 1 55 Sample Input 2 789 Sample Output 2 66066

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int rn,n,nt=0,i=0;
5     scanf("%d",&n);
6     do{
7         nt=n;rn=0;
8         while(n!=0)
9         {
10             rn=rn*10+n%10;
11             n=n/10;
12         }
13         n=nt+rn;
14         i++;
15     }
16     while(rn!=nt||i==1);
17     printf("%d",rn);
18     return 0;
19 }
```

	Input	Expected	Got	
✓	32	55	55	✓
✓	789	66066	66066	✓

Passed all tests! ✓

Question 3

Correct

Marked out of 7.00

Flag question

A number is considered lucky if it contains either 3 or 4 or 3 and 4 both in it. Write a program to print the nth lucky number. Example, 1st lucky number is 3, and 2nd lucky number is 4 and 3rd lucky number is 33 and 4th lucky number is 34 and so on. Note that 13, 40 etc., are not lucky as they have other numbers in it.

The program should accept a number 'n' as input and display



Passed all tests! ✓

Question **3**

Correct

Marked out of 7.00

Flag question

A number is considered lucky if it contains either 3 or 4 or 3 and 4 both in it. Write a program to print the nth lucky number. Example, 1st lucky number is 3, and 2nd lucky number is 4 and 3rd lucky number is 33 and 4th lucky number is 34 and so on. Note that 13, 40 etc., are not lucky as they have other numbers in it.

The program should accept a number 'n' as input and display the nth lucky number as output.

Sample Input 1:

3

Sample Output 1:

33

Explanation:

Here the lucky numbers are 3, 4, 33, 34., and the 3rd lucky number is 33.

Sample Input 2:

34

Sample Output 2:

33344

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n=1,i=0,nt,co=0,e;
5     scanf("%d",&e);
6     while(i<e)
7     {
8         nt=n;
9         while(nt!=0)
10        {
11            co=0;
12            if(nt%10!=3&&nt%10!=4)
13            {
14                co=1;
15                break;
16            }
17            nt=nt/10;
18        }
19        if(co==0)
20            i++;
21            n++;
22            }printf("%d",--n);
23    }
24 }
```

	Input	Expected	Got	
✓	34	33344	33344	✓

Passed all tests! ✓

Finish review



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Status	Finished
Started	Monday, 23 December 2024, 5:33 PM
Completed	Saturday, 21 December 2024, 11:15 AM
Duration	2 days 6 hours

Question **1**

Correct

Marked out of 3.00

Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[i] - A[j] = k, i \neq j$.

Input Format

- 1. First line is number of test cases T. Following T lines contain:
- 2. N, followed by N integers of the array
- 3. The non-negative integer k

Output format

Print 1 if such a pair exists and 0 if it doesn't.

Example

Input:

1
3 1 3 5
4

Output:

1

Input:

1
3 1 3 5
99

Output:

0

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main(){
3     int t;
4     scanf("%d",&t);
5     while(t--){
6         int n;
7         scanf("%d",&n);
8         int a[n];
9         for(int i=0;i<n;i++){
10             scanf("%d",&a[i]);
11         }
12         int k;
13         scanf("%d",&k);
14         int flag=0;
15         for (int i=0;i<n;i++){
16             for (int j=i+1;j<n;j++){
17                 if(a[i]-a[j]==k||a[j]-a[i]==k){
18                     flag=1;break;}
19             }
20             if(flag)break;}
21         printf("%d\n",flag);
```




REC-CIS

0

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main(){
3     int t;
4     scanf("%d",&t);
5     while(t--){
6         int n;
7         scanf("%d",&n);
8         int a[n];
9         for(int i=0;i<n;i++){
10             scanf("%d",&a[i]);
11         }
12         int k;
13         scanf("%d",&k);
14         int flag=0;
15         for (int i=0;i<n;i++){
16             for (int j=i+1;j<n;j++){
17                 if(a[i]-a[j]==k||a[j]-a[i]==k){
18                     flag=1;break;}
19             }
20         }
21         printf("%d\n",flag);
22     }
23 }
```

	Input	Expected	Got	
✓	1 3 1 3 5 4	1	1	✓
✓	1 3 1 3 5 99	0	0	✓

Passed all tests! ✓

Question 2

Correct

Marked out of 5.00

Flag question

Sam loves chocolates and starts buying them on the 1st day of the year. Each day of the year, x , is numbered from 1 to Y . On days when x is odd, Sam will buy x chocolates; on days when x is even, Sam will not purchase any chocolates.

Complete the code in the editor so that for each day N_i (where $1 \leq x \leq N \leq Y$) in array `arr`, the number of chocolates Sam purchased (during days 1 through N) is printed on a new line. This is a function-only challenge, so input is handled for you by the locked stub code in the editor.

Input Format

The program takes an array of integers as a parameter.

The locked code in the editor handles reading the following input from `stdin`, assembling it into an array of integers (`arr`), and calling `calculate(arr)`.

The first line of input contains an integer, T (the number of test cases). Each line i of the T subsequent lines describes the i th test case as an integer, N_i (the number of days).

Constraints

$$1 \leq T \leq 2 \times 10^5$$

$$1 \leq N \leq 2 \times 10^6$$

$$1 \leq x \leq N \leq Y$$

Output Format

For each test case, T_i in `arr`, your `calculate` method should



REC-CIS

Test Case 1: N = 2

Sam buys 1 chocolate on day 1 and 0 on day 2. This gives us a total of 1 chocolate. Thus, we print 1 on a new line.

Test Case 2: N = 3

Sam buys 1 chocolate on day 1, 0 on day 2, and 3 on day 3. This gives us a total of 4 chocolates. Thus, we print 4 on a new line.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main(){
3     int t;
4     scanf("%d",&t);
5     while(t--){
6         int n,c=0;
7         scanf("%d",&n);
8         for(int i=0;i<=n;i++){
9             if(i%2!=0)c=c+i;
10        }
11        printf("%d\n",c);
12    }
13 }
```

	Input	Expected	Got	
✓	3	1	1	✓
	1	1	1	
	2	4	4	
	3			
✓	10	1296	1296	✓
	71	2500	2500	
	100	1849	1849	
	86	729	729	
	54	400	400	
	40	25	25	
	9	1521	1521	
	77	25	25	
	9	49	49	
	13	2401	2401	
	98			

Passed all tests! ✓

Question 3

Correct

Marked out of 7.00

Flag question

The number of goals achieved by two football teams in matches in a league is given in the form of two lists. Consider:

- Football team A, has played three matches, and has scored { 1 , 2 , 3 } goals in each match respectively.
- Football team B, has played two matches, and has scored { 2 , 4 } goals in each match respectively.
- Your task is to compute, for each match of team B, the total number of matches of team A, where team A has scored less than or equal to the number of goals scored by team B in that match.
- In the above case:
- For 2 goals scored by team B in its first match, team A has 2 matches with scores 1 and 2.
- For 4 goals scored by team B in its second match, team A has 3 matches with scores 1, 2 and 3.

Hence, the answer: {2, 3}.



3
1
7
8

Sample Output 1

1
0
3
4

Explanation 1

We are given, n = 5, nums = [2, 10, 5, 4, 8], m = 4, and maxes = [3, 1, 7, 8].

- 1. For maxes[0] = 3, we have 1 element in nums (nums[0] = 2) that is ≤ maxes[0].
- 2. For maxes[1] = 1, there are 0 elements in nums that are ≤ maxes[1].
- 3. For maxes[2] = 7, we have 3 elements in nums (nums[0] = 2, nums[2] = 5, and nums[3] = 4) that are ≤ maxes[2].
- 4. For maxes[3] = 8, we have 4 elements in nums (nums[0] = 2, nums[2] = 5, nums[3] = 4, and nums[4] = 8) that are ≤ maxes[3].

Thus, the function returns the array [1, 0, 3, 4] as the answer.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int s1,s2,ans;
5     scanf("%d",&s1);
6     int ta[s1];
7     for(int i=0;i<s1;i++)
8         scanf("%d",&ta[i]);
9     scanf("%d",&s2);
10    int tb[s2];
11    for(int i=0;i<s2;i++)
12        scanf("%d",&tb[i]);
13    for(int j=0;j<s2;j++)
14    {
15        ans=0;
16        for(int i=0;i<s1;i++){
17            if(tb[j]>=ta[i])
18                ans++;
19        }
20        printf("%d\n",ans);
21    }
22 }
```

	Input	Expected	Got	
✓	4 1 4 2 4 2 3 5	2 4	2 4	✓
✓	5 2 10 5 4 8 4 3 1 7 8	1 0 3 4	1 0 3 4	✓

Passed all tests! ✓



GE23131-Programming Using C-2024

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Finish review

Question 1

Correct

Marked out of 5.00

Flag question

Status	Finished
Started	Monday, 23 December 2024, 5:33 PM
Completed	Monday, 23 December 2024, 11:08 AM
Duration	6 hours 24 mins

Sunny and Johnny like to pool their money and go to the ice cream parlor. Johnny never buys the same flavor that Sunny does. The only other rule they have is that they spend all of their money.

Given a list of prices for the flavors of ice cream, select the two that will cost all of the money they have.

For example, they have $m = 6$ to spend and there are flavors costing $cost = [1, 2, 3, 4, 5, 6]$. The two flavors costing 1 and 5 meet the criteria. Using 1-based indexing, they are at indices 1 and 4.

Function Description

Complete the code in the editor below. It should return an array containing the indices of the prices of the two flavors they buy.

It has the following:

- m : an integer denoting the amount of money they have to spend
- $cost$: an integer array denoting the cost of each flavor of ice cream

Input Format

The first line contains an integer, t , denoting the number of trips to the ice cream parlor. The next t sets of lines each describe a visit. Each trip is described as follows:

- The integer m , the amount of money they have pooled.
- The integer n , the number of flavors offered at the time.
- n space-separated integers denoting the cost of each flavor: $cost[1], cost[2], \dots, cost[n]$.

Note: The index within the cost array represents the flavor of the ice cream purchased.

Constraints

- $1 \leq t \leq 50$
- $2 \leq m \leq 10^4$
- $2 \leq n \leq 10^4$
- $1 \leq cost[i] \leq 10^4, \forall i \in [1, n]$
- There will always be a unique solution.

Output Format

For each test case, print two space-separated integers denoting the indices of the two flavors purchased, in ascending order.

Sample Input



REC-CIS

1 4
1 2

Explanation

Sunny and Johnny make the following two trips to the parlor:

- 1. The first time, they pool together $m = 4$ dollars. Of the five flavors available that day, flavors **1** and **4** have a total cost of $1 + 3 = 4$.
- 2. The second time, they pool together $m = 4$ dollars. TOf the four flavors available that day, flavors **1** and **2** have a total cost of $2 + 2 = 4$.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main(){
3     int t,m,n,c=0;
4     scanf("%d",&t);
5     for(int i=0;i<t;i++){
6         c=0;
7         scanf("%d\n %d",&m,&n);
8         int arr[n];
9         for(int j=0;j<n;j++){
10             scanf("%d",&arr[j]);
11         }
12         for(int a=0;a<n-1;a++){
13             for(int b=a+1;b<n;b++){
14                 if(arr[a]+arr[b]==m){
15                     printf("%d %d\n",a+1,
16                         c=1;break;
17                 }
18             }
19             if(c==1)break;
20         }
21     }
22     return 0;
23 }
24
```

	Input	Expected	Got	
✓	2	1 4	1 4	✓
	4	1 2	1 2	
	5			
	1 4 5 3 2			
	4			
	4			
	2 2 4 3			

Passed all tests! ✓

Question 2

Correct

Marked out of 5.00

Flag question

Numeros the Artist had two lists that were permutations of one another. He was very proud. Unfortunately, while transporting them from one exhibition to another, some numbers were lost out of the first list. Can you find the missing numbers?

As an example, the array with some numbers missing, $arr = [7, 2, 5, 3, 5, 3]$. The original array of numbers $brr = [7, 2, 5, 4, 6, 3, 5, 3]$. The numbers missing are $[4, 6]$.

Notes

- If a number occurs multiple times in the lists, you must ensure that the frequency of that number in both lists is the same. If that is not the case, then it is also a missing number.
- You have to print all the missing numbers in ascending order.
- Print each missing number once, even if it is missing

10
203 204 205 206 207 208 203 204 205 206
13
203 204 204 205 206 207 205 208 203 206 205 206 204

Sample Output

204 205 206

Explanation

204 is present in both arrays. Its frequency in **arr** is **2**, while its frequency in **brr** is **3**. Similarly, **205** and **206** occur twice in **arr**, but three times in **brr**. The rest of the numbers have the same frequencies in both lists.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,m,c,c1=0,c0;
5     scanf("%d",&n);
6     int arr[n];
7     for(int a=0;a<n;a++)
8         scanf("%d",&arr[a]);
9     scanf("%d",&m);
10    int brr[m],ans[m];
11    for(int b=0;b<m;b++)
12        scanf("%d",&brr[b]);
13    for(int j=0;j<m;j++)
14    {
15        c=0;
16        for(int i=0;i<n;i++)
17        {
18            if(arr[i]==brr[j])
19            {
20                c=1;
21                arr[i]=1;
22                break;
23            }
24        }
25        if(c==0)
26        {
27            ans[c1]=brr[j];
28            c1++;
29        }
30    }
31    for(int a=0;a<c1;a++)
32    {
33        c0=0;
34        for(int b=0;b<c1;b++)
35        {
36            if(ans[b]<ans[a])
37                c0++;
38        }
39        int temp=ans[a];
40        ans[a]=ans[c0];
41        ans[c0]=temp;
42    }
43    for(int i=0;i<c1;i++)
44        printf("%d ",ans[i]);
45    return 0;
46 }
```

	Input
✓	10 203 204 205 206 207 208 203 204 205 206 13 203 204 204 205 206 207 205 208 203 206 205
Passed all tests! ✓	

Question **3**

Correct

Marked out of 5.00

Flag question

Watson gives Sherlock an array of integers. His challenge is to find an element of the array such that the sum of all elements to the left is equal to the sum of all elements to the right. For instance, given the array **arr = [5, 6, 8, 11]**, **8** is between two subarrays that sum to **11**. If your starting array is **[1]**, that element satisfies the rule as left and right sum to **0**.

You will be given arrays of integers and must determine

For the first test case, no such index exists.

For the second test case, $arr[0] + arr[1] = arr[3]$, therefore index **2** satisfies the given conditions.

Sample Input 1

3
5
1 1 4 1 1
4
2 0 0 0
4
0 0 2 0

Sample Output 1

YES
YES
YES

Explanation 1

In the first test case, $arr[2] = 4$ is between two subarrays summing to **2**.

In the second case, $arr[0] = 2$ is between two subarrays summing to **0**.

In the third case, $arr[2] = 2$ is between two subarrays summing to **0**.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int t,n,Is,rs,m;
5     scanf("%d",&t);
6     for(int i=0;i<t;i++){
7         Is=0;
8         rs=0;
9         scanf("%d",&n);
10        int arr[n];
11        for(int j=0;j<n;j++)
12            scanf("%d",&arr[j]);
13        m=n/2;
14        if(arr[m]==0)
15        {
16            for(m=0;arr[m]==0&&m<n;m++);
17        }
18        for(int j=0;j<=m;j++)
19            Is=Is+arr[j];
20        for(int j=m;j<n;j++)
21            rs=rs+arr[j];
22        printf("%s\n",(Is==rs)?"YES":"NO");
23    }
24    return 0;
25 }
```

	Input	Expected	Got	
✓	3 5 1 1 4 1 1 4 2 0 0 0 4 0 0 2 0	YES YES YES	YES YES YES	✓
✓	2 3 1 2 3 4 1 2 3 3	NO YES	NO YES	✓

Passed all tests! ✓



GE23131-Programming Using C-2024

Quiz navigation



Show one page at a time

Finish review

Status	Finished
Started	Monday, 13 January 2025, 11:44 AM
Completed	Monday, 13 January 2025, 12:28 PM
Duration	43 mins 55 secs

Question **1**

Correct

Marked out of 1.00

Flag question

Coders here is a simple task for you, you have given an array of size **N** and an integer **M**.

Your task is to calculate the **difference between maximum sum and minimum sum of N-M** elements of the given array.

Constraints:

- 1<=t<=10
- 1<=n<=1000
- 1<=a[i]<=1000

Input:

First line contains an integer **T** denoting the number of testcases.
First line of every testcase contains two integer **N** and **M**.
Next line contains **N** space separated integers denoting the elements of array

Output:

For every test case print your answer in new line

SAMPLE INPUT

```
1
5 1
1 2 3 4 5
```

SAMPLE OUTPUT

```
4
```

Explanation

M is 1 and N is 5 so you have to calculate maximum and minimum sum using (5-1 =) 4 elements.
Maximum sum using the 4 elements would be (2+3+4+5=)14.
Minimum sum using the 4 elements would be (1+2+3+4=)10.
Difference will be 14-10=4.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int t;
5     scanf("%d",&t);
6     while(t-->0)
7     {
8         int n,m,d,min,temp;
9         scanf("%d %d",&n,&m);
10        d=n-m;
11        int arr[n];
```